Foundation of Blockchain Technology (IS501)
Pre-Reading Material – Lecture 13

Ethereum Wallets

Estimated Reading Time: 15–20 minutes

Why This Topic Matters

To interact with the Ethereum blockchain, users need an **Ethereum account and its cryptographic keys**. The most important of these is the **private key**, which provides access to the assets and activities associated with the account.

An **Ethereum wallet** provides a way to securely manage these keys and use them to interact with the blockchain.

Understanding wallets is therefore essential before working with Ethereum applications, transactions, and smart contracts.

Learning Outcomes

After completing this pre-reading, you should be able to:

- Explain what an **Ethereum wallet** is.
- Understand the role of **public and private keys**.
- Explain why protecting the **private key** is essential.
- Differentiate between **hot and cold wallets**.
- Identify different types of Ethereum wallets.
- Compare wallets based on **security and convenience**.
- Understand the basic purpose of **software, hardware, and paper wallets**.

Activate Your Prior Knowledge

1. Ethereum Accounts

You have learned that Ethereum uses accounts to interact with the blockchain.

Think:

If an Ethereum account has cryptographic keys, where should those keys be stored?

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2. Private Keys

The private key is required to authorize actions involving an account.

Think:

What could happen if someone else obtains your private key?

3. Security vs. Convenience

Some technologies are easier to access but may introduce greater security risks.

Think:

Would you prefer a wallet that is extremely convenient or one that provides maximum security?

Prerequisite Concepts

You should be familiar with:

- Ethereum
- Blockchain
- Ethereum accounts
- Public and private keys
- Cryptocurrency
- Basic cryptography
- Blockchain transactions

Core Concepts

1. What is an Ethereum Wallet?

An **Ethereum wallet** is a mechanism used to **store and manage the cryptographic keys** associated with an Ethereum account.

The wallet does **not actually store the cryptocurrency on the device**.

Instead:

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Blockchain → Stores the assets and transaction records

Wallet → Stores/manages the keys that provide access to them

The **private key** is particularly important because it allows the user to interact with the assets associated with the account.

Therefore:

Your assets are recorded on the blockchain; your wallet manages the keys that give you access to them.

2. Public Key and Private Key

Ethereum accounts use cryptographic keys.

Private Key

The private key must be kept **secret and secure**.

It is used to authorize actions from the account.

Public Key

The public key can be shared and is related to identifying the account.

A simplified view is:

Private Key → Securely controlled by the user

Public Key → Can be shared

Important

If you lose access to your private key, you may lose access to the assets associated with that account.

If someone else obtains your private key, they may be able to control the associated assets.

3. Security vs. Convenience

When choosing a wallet, two major factors need to be considered:

- **Security**
- **Convenience**

Generally:

More convenience ↔ potentially less security

More security ↔ potentially less convenience

Therefore, choosing a wallet involves balancing these two factors according to the user's needs.

4. Hot Wallets

A **hot wallet** stores or manages keys while being connected to the internet.

Advantages

- Easy to access
- Convenient for frequent transactions
- Can be accessed through devices such as computers or smartphones

Disadvantage

Because the keys are accessible online, they may have greater exposure to online attacks.

Think:

Why might an internet-connected wallet be more convenient but potentially more vulnerable?

5. Cold Wallets

A **cold wallet** keeps the keys **offline**.

Examples include:

- Hardware wallets

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- Paper wallets

Advantages

- Less exposure to online attacks
- Suitable when security is a major priority

Disadvantages

- Less convenient
- Physical loss or damage can result in loss of access if proper backups are not maintained

Therefore:

Hot Wallet → Online → More convenient

Cold Wallet → Offline → Generally more isolated from online threats

6. Types of Ethereum Wallets

Ethereum wallets can be implemented in different forms.

Software Wallets

Software wallets use applications to manage keys.

Common forms include:

- **Web wallet**
- **Desktop wallet**
- **Mobile wallet**

Hardware Wallets

A hardware wallet is a **physical device** designed to securely manage private keys.

Paper Wallets

A paper wallet involves keeping the keys or relevant wallet information **offline on paper**.

Each type involves different trade-offs between security, convenience, accessibility, and backup requirements.

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7. Web Wallets

A **web wallet** is accessed through a web browser.

It provides convenient access to the wallet from internet-connected devices.

However, users must pay particular attention to security because online wallets can be exposed to threats such as phishing and account compromise.

8. Desktop Wallets

A **desktop wallet** is software installed on a computer.

The wallet manages the user's keys locally through the installed application.

It can provide greater control over key management compared with a fully online service, depending on how the wallet is designed and used.

9. Mobile Wallets

A **mobile wallet** is a wallet application running on a smartphone or tablet.

Its major advantage is **portability and convenience**.

Mobile wallets can also make transactions convenient through features such as **QR codes**.

10. Hardware Wallets

A **hardware wallet** is a physical device designed to securely manage private keys.

The key idea is:

Private Key → Hardware Device

The device can be connected to a computer when the user needs to interact with the blockchain.

Hardware wallets are generally designed to provide strong security while still allowing users to access their assets when required.

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11. Paper Wallets

A **paper wallet** involves keeping the relevant key information offline on paper.

Advantage

The key is not directly exposed to online attacks.

Disadvantage

If the paper is lost or damaged and there is no reliable backup, access to the associated assets may be lost.

Therefore, physical security and backup are extremely important.

Guided Reading Questions

1. What is an **Ethereum wallet**?
2. Does the wallet actually store cryptocurrency on the device?
3. What is the role of the **private key**?
4. Why must the private key be protected?
5. What is the difference between a **hot wallet** and a **cold wallet**?
6. What are the major types of Ethereum wallets?
7. What is a **web wallet**?
8. What is a **desktop wallet**?
9. What is a **mobile wallet**?
10. What is a **hardware wallet**?
11. What is a **paper wallet**?
12. Why is there a trade-off between security and convenience?

Reflection Activity

Write 75–100 words explaining why the private key is more important than the wallet itself when protecting access to Ethereum assets. Also explain why losing the private key can be a serious problem.

Self-Assessment

1. An Ethereum wallet primarily manages the user's ___________.
2. The most important key that must be kept secret is the ___________ key.

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3. A wallet connected to the internet is generally called a __________ wallet.
4. A wallet that keeps keys offline is called a __________ wallet.
5. A wallet accessed through a browser is called a __________ wallet.
6. A wallet installed on a computer is called a __________ wallet.
7. A wallet application running on a smartphone is called a __________ wallet.
8. A physical device used to securely manage keys is called a __________ wallet.
9. A wallet that stores key information offline on paper is called a __________ wallet.
10. The two important considerations when choosing a wallet are security and __________.

Key Takeaway

An **Ethereum wallet does not store cryptocurrency itself**; it manages the **cryptographic keys** that allow users to interact with assets recorded on the Ethereum blockchain.

The most important concept is:

Ethereum Blockchain → Stores assets and records

Wallet → Manages keys

Wallets can be broadly classified as:

Hot → Online → Convenient

Cold → Offline → More isolated from online threats

And they can take different forms:

Web • Desktop • Mobile • Hardware • Paper

The right wallet depends on the required balance between **security, convenience, and accessibility**.

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